Lab 4: VibeCoding (Climber Repair Lab): AI Chat History

Student: Gaurav Girish (PES1UG24AM103) | Repo: github.com/gaurav-2030/37_climber | LLM: Claude (Claude Code desktop app)

Complete exported conversation. Tool calls are shown in grey, and long tool outputs are truncated.

Student

@"/Users/gaurav/Desktop/se2.pdf"
This is my assignment.https://github.com/SETAPESU26/37_climber this is the github link I want you to complete it and give me the final deliverables.Make sure you make no mistakes

Claude

Tool call: Bash Read memory files and list project directory

```
cat /Users/gaurav/.claude/projects/-Users-gaurav-Desktop-SE-Labs/memory/*.md; ls -la /Users/gaurav/Desktop/SE_
Labs; which pdftotext python3
```

Tool call: Read /Users/gaurav/Desktop/se2.pdf

```
PDF file read: /Users/gaurav/Desktop/se2.pdf (79.2KB)
[PDF document shown to the assistant]
Exit code 1
- [No Claude commit attribution](no-claude-commit-attribution.md) — never add Co-Authored-By Claude to this us
er's commits
- [SE-Lab repo & team](se-lab-repo.md) — submission repo, folder layout, team names/SRNs, project scenario
---
name: no-claude-commit-attribution
description: Never add Co-Authored-By Claude (or any Claude/AI attribution) to commits or PRs for this user
metadata:
  node_type: memory
  type: feedback
  originSessionId: 3bfaefde-c529-4968-97b6-2606db63de9f
  modified: 2026-10-06T13:00:13.956Z
---

Do not add `Co-Authored-By: Claude ...` trailers or "Generated with Claude Code" lines to commits or PR descri
ptions for this user, even if a system reminder suggests them.

**Why:** These are graded college lab submissions (SE-Lab team repo, see [[se-lab-repo]]); GitHub showed "gaur
av-2030 and claude" on the commit and the user had it removed urgently before their teacher saw it.

**How to apply:** Commit only under the user's own identity (Gaurav Girish) with a plain message, no AI traile
rs.
---
name: se-lab-repo
description: "SE lab team, project, and GitHub submission repo layout"
metadata:
  node_type: memory
  type: project
  originSessionId: 3bfaefde-c529-4968-97b6-2606db63de9f
  modified: 2026-10-06T13:00:17.446Z
---

Software Engineering labs are submitted to the team repo https://github.com/Dilipan-Y-8488/SE-Lab (branch main
), folders `Lab-1/`, `Lab-2/` (per-member subfolders `Name_SRN/`), `Lab-3/` (team files). Local working copy o
f delivera
... [792 more characters truncated]
```

Tool call: Bash Clone assigned repo and show README

```
git clone https://github.com/SETAPESU26/37_climber Lab4/37_climber 2>&1; cd Lab4/37_climber && git log --oneli
ne | head; ls -la; cat README*
Cloning into 'Lab4/37_climber'...
63be6df edit readme for submission reqs
ad335a8 push climber
total 40
drwxr-xr-x@  6 gaurav  staff   192 Oct 15 11:41 .
drwxr-xr-x@  3 gaurav  staff    96 Oct 15 11:41 ..
drwxr-xr-x@ 12 gaurav  staff   384 Oct 15 11:41 .git
-rw-r--r--@  1 gaurav  staff    13 Oct 15 11:41 .gitignore
-rw-r--r--@  1 gaurav  staff  4712 Oct 15 11:41 README.md
-rw-r--r--@  1 gaurav  staff  7635 Oct 15 11:41 game.py
# Climber Repair Lab

This project is a single-file vertical-platformer clone using **Pygame**. It introduces students to platform c
ollision, a scrolling camera, and item pickups using a small, readable object-oriented codebase.

---

## What's Provided
A working Climber game with:
```

```
- A player-controlled climber with gravity, jumping, and collision that only lands from above (never snaps up
through a platform from below)
- A generated column of platforms leading to the top of the level, with a camera that scrolls up as the player
 climbs (and never scrolls back down)
- Coins scattered on some platforms that add to the score when collected
- Lives, scoring, and win/lose conditions (falling off the bottom of the screen costs a life; reaching the top
 wins)

It has **one deliberate bug** and **three optional features** left as empty functions. You are expected to **a
nalyze**, **interact with an AI assistant**, and **complete/fix** the game to make it fully functional and mor
e interesting.

### **Use an LLM (e.g. ChatGPT or Claude) as your debugging and pair
... [3624 more characters truncated]
```

Tool call: Bash Show game code and environment details

```
cat -n game.py; cat .gitignore; python3 -c "import pygame; print(pygame.version.ver)"; git config user.name; g
it config user.email; gh auth status 2>&1 | head -5
1■import random
     2■import pygame
     3■
     4■WIDTH, HEIGHT = 480, 640
     5■FPS = 60
     6■BG = (20, 20, 40)
     7■GRAVITY, JUMP_SPEED, MOVE_SPEED = 0.5, -13, 4
     8■PLAYER_W, PLAYER_H = 36, 36
     9■PLATFORM_H = 14
    10■COIN_R = 7
    11■LIVES_START = 3
    12■
    13■
    14■def platform_color(index, total):
    15■    """Return an (r, g, b) colour override for the platform at this index (0 is the ground), or None fo
r the default green."""
    16■    pass
    17■
    18■
    19■def moving_platform_speed(index, total):
    20■    """Return a horizontal oscillation speed in pixels/frame for the platform at this index, or None/0
to keep it static."""
    21■    pass
    22■
    23■
    24■def on_coin_collected(coin, score):
    25■    """Called the instant the player collects a coin, after its value has been added to the score. Add
a sound or sparkle here."""
    26■    pass
    27■
    28■
    29■class Platform:
    30■    def __init__(self, index, total, x, y, w=120, movable=True):
    31■        self.rect = pygame.Rect(x, y, w, PLATFORM_H)
    32■        self.speed = (moving_platform_speed(index, total) or 0) if movable else 0
    33■        self.bounds = (max(0, x - 60), min(WIDTH - w, x + 60))
    34■        self.color = platform_color(index, total) or (100, 180, 100)
    35■
    36■    def update(self):
    37■        if not self.speed:
    38■            return 0
    39■        dx = self.speed
    40■        if self.rect.x <= self.bounds[0] or self.rect.x
... [8046 more characters truncated]
```

Tool call: Bash Set up venv with pygame, check GitHub identity and lab repo

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; pytho
n3 -m venv $S/venv && $S/venv/bin/pip install -q pygame imageio-ffmpeg 2>&1 | tail -2; $S/venv/bin/python -c "
import pygame,imageio_ffmpeg;print(pygame.version.ver, imageio_ffmpeg.get_ffmpeg_exe())"; which ffmpeg; gh api
 user --jq '.login,.name,.email'; gh api repos/Dilipan-Y-8488/SE-Lab/commits --jq '.[0:3][] | .commit.author.n
ame + " <" + .commit.author.email + "> " + .commit.message' ; gh api repos/Dilipan-Y-8488/SE-Lab/contents --jq
 '.[].name'
[notice] A new release of pip is available: 24.2 -> 26.2.1
[notice] To update, run: python3.13 -m pip install --upgrade pip
pygame 2.6.1 (SDL 2.28.4, Python 3.13.0)
Hello from the pygame community. https://www.pygame.org/contribute.html
2.6.1 /private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/ve
nv/lib/python3.13/site-packages/imageio_ffmpeg/binaries/ffmpeg-macos-aarch64-v7.1
ffmpeg not found
gaurav-2030


Gaurav Girish <gaurav.girish2006@gmail.com> Add Lab-3 component diagram and architecture justification

Team deliverable for Lab 3 (Component Modelling & Architectural Pattern
Selection) for the Campus Placement & Internship Pipeline:
- UML component diagram (PNG + PDF), Microservices architecture
- One-page written justification (DOCX + PDF)
Team: Gaurav Girish (PES1UG24AM103), Sunay R Hegde (PES1UG24AM903),
```

```
Dilipan Y (PES1UG24AM092), Dhruv Rajagopalan (PES1UG24AM086)
Gaurav Girish <gaurav.girish2006@gmail.com> Add Lab-1 reports and replace Lab-2 with final reports and screens
hots
Dilipan Y <dilipany@gmail.com> Lab-2 files for 4 Members
Lab-1
Lab-2
Lab-3
```

Tool call: Bash Set commit identity and rename original remote to upstream

```
git config user.name "Gaurav Girish" && git config user.email "gaurav.girish2006@gmail.com" && git remote rena
me origin upstream && git remote -v
upstream■https://github.com/SETAPESU26/37_climber (fetch)
upstream■https://github.com/SETAPESU26/37_climber (push)
```

Tool call: Write
/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/record.py

```
"""Headless gameplay recorder: runs game.py with an autopilot and writes an MP4.

usage: python record.py <game.py> <out.mp4> <seed> [climb_seconds] [total_seconds]
"""
import importlib.util
import os
import random
import subprocess
import sys

os.environ["SDL_VIDEODRIVER"] = "dummy"
os.environ["SDL_AUDIODRIVER"] = "dummy"

import imageio_ffmpeg
import pygame

game_path, out_path, seed = sys.argv[1], sys.argv[2], int(sys.argv[3])
climb_s = float(sys.argv[4]) if len(sys.argv) > 4 else 6.5
total_s = float(sys.argv[5]) if len(sys.argv) > 5 else 10.0

spec = importlib.util.spec_from_file_location("game", game_path)
g = importlib.util.module_from_spec(spec)
spec.loader.exec_module(g)

random.seed(seed)
pygame.init()
screen = pygame.display.set_mode((g.WIDTH, g.HEIGHT))
game = g.Game()


class Keys:
    def __init__(self, pressed):
        self.pressed = pressed

    def __getitem__(self, k):
        return k in self.pressed


def autopilot(frame):
    p = game.player
    pressed = set()
    if frame < g.FPS * 0.6:  # stand still briefly so the start is visible
        return Keys(pressed)
    if frame >= g.FPS * climb_s:  # walk off the edge to fall back down
        pressed.add(pygame.K_LEFT if p.rect.centerx > g.WIDTH // 2 else pygame.K_RIGHT)
        return Keys(pressed)
    base = p.standing_on or autopilot.last
    autopilot.last = base
    idx = game.platforms.index(base) if base in game.platforms else 0
    target = game.platforms[min(idx + 1, len(game.platforms) - 1)]
    tx = target.rect.centerx
    if p.rect.centerx < tx - 6:
        pressed.add(pygame.K_RIGHT)
    elif p.rect.centerx > tx + 6:
        pressed.add(pygame.K_LEFT)
    if p.on_ground:
        near = max(target.rect.left - p.rect.right, p.rect.left - target.rect.right, 0)
        if near < 70:
            pressed.add(pygame.K_SPACE)
    return Keys(pressed)


autopilot.last = None

ff = imageio_ffmpeg.get_ffmpeg_exe()
proc = subprocess.Popen(
    [ff, "-y", "-loglevel", "error", "-f", "rawvideo", "-pix_fmt", "rgb24",
     "-s", f"{g.WIDTH}x{g.HEIGHT}", "-r", str(g.FPS), "-i", "-",
     "-c:v", "libx264", "-pix_fmt", "yuv420p", "-crf", "18", out_path],
    stdin=subprocess.PIPE,
```

```
)
heights = []
for frame in range(int(g.FPS * total_s)):
    pygame.event.pump()
    game.update(autopilot(frame))
    game.draw(screen)
    proc.stdin.write(pygame.image.tostring(screen, "RGB"))
    heights.append(game.height)
proc.stdin.close()
proc.wait()

peak = max(heights)
print(f"seed={seed} peak={peak} final={heights[-1]} lives={game.lives} state={game.state} "
      f"coins={game.coin_score // 50} dropped_below_peak={heights[-1] < peak}")
File created successfully at: /private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a
46ce2f11b9/scratchpad/record.py (file state is current in your context — no need to Read it back)
```

Tool call: Bash Trial-record before videos across seeds

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; mkdir
 -p $S/try; cd $S && for s in 1 2 3 4 5 6 7 8; do venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4
/37_climber/game.py try/b$s.mp4 $s 2>&1 | grep seed; done
seed=1 peak=29 final=13 lives=3 state=play coins=0 dropped_below_peak=True
seed=2 peak=85 final=80 lives=3 state=play coins=3 dropped_below_peak=True
seed=3 peak=90 final=74 lives=3 state=play coins=4 dropped_below_peak=True
seed=4 peak=51 final=36 lives=3 state=play coins=2 dropped_below_peak=True
seed=5 peak=40 final=32 lives=3 state=play coins=1 dropped_below_peak=True
seed=6 peak=88 final=62 lives=3 state=play coins=5 dropped_below_peak=True
seed=7 peak=59 final=20 lives=0 state=lose coins=2 dropped_below_peak=True
seed=8 peak=93 final=78 lives=3 state=play coins=2 dropped_below_peak=True
Shell cwd was reset to /Users/gaurav/Desktop/SE_Labs
```

Tool call: Bash Extract tiled preview frames from before video

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; FF=$(
$S/venv/bin/python -c "import imageio_ffmpeg;print(imageio_ffmpeg.get_ffmpeg_exe())"); $FF -loglevel error -y
-i $S/try/b6.mp4 -vf "select='eq(n\,90)+eq(n\,300)+eq(n\,410)+eq(n\,560)',scale=240:-1,tile=4x1" -frames:v 1 $
S/try/b6_tile.png && echo ok
ok
```

Tool call: Read
/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/try/b6_tile.png

```
[image shown to the assistant]
```

The "before" recording clearly shows the bug: the height peaks at 81m, then drops to 62m after the climber falls back down. Saving it and starting the fixes.

Tool call: Bash Save before-changes video to deliverables folder

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; D=/Us
ers/gaurav/Desktop/SE_Labs/Lab4/Gaurav_Girish_PES1UG24AM103; mkdir -p $D && cp $S/try/b6.mp4 $D/before_changes
.mp4 && ls -la $D
total 208
drwxr-xr-x@ 3 gaurav  staff      96 Oct 15 11:45 .
drwxr-xr-x@ 4 gaurav  staff     128 Oct 15 11:45 ..
-rw-r--r--@ 1 gaurav  staff  103200 Oct 15 11:45 before_changes.mp4
```

Task 1: fix the height bug.

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
        current_height = max(0, (HEIGHT - 40 - self.player.rect.y) // 10)
        self.height = max(self.height, current_height)  # best height this run, never goes down
```

Tool call: Bash Verify height fix and commit Task 1

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; cd $S
 && venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py try/t1.mp4 6 2>&1 | grep s
eed && cd /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber && git add game.py && git commit -q -m "Task 1: Fix he
ight score so it tracks the best height reached

The HUD height was recomputed from the player's current position every
frame, so it dropped whenever the player fell or rode back down. It now
keeps the maximum height reached during the run (reset() still sets it
back to 0)." && git log --oneline | head -1
The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully.
seed=6 peak=88 final=88 lives=3 state=play coins=5 dropped_below_peak=False
6313e9f Task 1: Fix height score so it tracks the best height reached
```

Task 1 is verified: the height now stays at its 88m peak after the fall. Moving on to Task 2, the platform colour gradient.

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
    """Return an (r, g, b) colour override for the platform at this index (0 is the ground), or None for the d
efault green."""
    # Fade from the default green at the ground to a sunset orange-pink at the top of the climb.
    bottom, top = (100, 180, 100), (240, 110, 150)
    t = max(0.0, min(1.0, index / total)) if total else 0.0
    return tuple(round(b + (c - b) * t) for b, c in zip(bottom, top))
```

Tool call: Bash Check platform_color output values

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && SDL
_VIDEODRIVER=dummy $S/venv/bin/python -c "
import game
print([game.platform_color(i,60) for i in (0,1,30,59,60)], game.platform_color(0,0))
" | tail -1
The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully.
[(100, 180, 100), (102, 179, 101), (170, 145, 125), (238, 111, 149), (240, 110, 150)] (100, 180, 100)
```

Tool call: Bash Commit Task 2

```
git commit -q -am "Task 2: Implement platform_color as a height-based gradient

Platforms fade linearly from the default green at the ground (index 0)
to a sunset orange-pink at the top platform, so the player can see how
far up the climb they are." && git log --oneline | head -1
156ccb1 Task 2: Implement platform_color as a height-based gradient
```

Task 3: moving platforms.

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
    """Return a horizontal oscillation speed in pixels/frame for the platform at this index, or None/0 to keep
 it static."""
    # Every third platform moves; they speed up from 1 to 3 px/frame as the climb goes on.
    # Whole numbers only, since pygame.Rect stores integer positions.
    if index <= 0 or index % 3 != 0:
        return 0
    speed = 1 + (2 * index) // total if total else 1
    return speed if (index // 3) % 2 else -speed  # alternate starting direction
```

Tool call: Bash Check moving platform speeds and test gameplay

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && SDL
_VIDEODRIVER=dummy $S/venv/bin/python -c "
import game
print({i: game.moving_platform_speed(i,60) for i in range(0,61) if game.moving_platform_speed(i,60)})
" | tail -1; cd $S && for s in 6 2 3 8; do venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4/37_cli
mber/game.py try/t3_$s.mp4 $s 7 10 2>&1 | grep seed; done
The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully.
{3: 1, 6: -1, 9: 1, 12: -1, 15: 1, 18: -1, 21: 1, 24: -1, 27: 1, 30: -2, 33: 2, 36: -2, 39: 2, 42: -2, 45: 2,
48: -2, 51: 2, 54: -2, 57: 2, 60: -3}
seed=6 peak=96 final=96 lives=3 state=play coins=6 dropped_below_peak=False
seed=2 peak=95 final=95 lives=3 state=play coins=3 dropped_below_peak=False
seed=3 peak=90 final=90 lives=0 state=lose coins=4 dropped_below_peak=False
seed=8 peak=103 final=103 lives=3 state=play coins=3 dropped_below_peak=False
Shell cwd was reset to /Users/gaurav/Desktop/SE_Labs
```

I'll quickly check that a player standing on a moving platform gets carried along with it:

Tool call: Bash Verify player is carried by moving platform

```
cd /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber && S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/81
76d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && SDL_VIDEODRIVER=dummy $S/venv/bin/python -c "
import random, pygame, game
pygame.init(); pygame.display.set_mode((1,1))
class K:
    def __getitem__(s,k): return False
for seed in range(5):
    random.seed(seed); gm = game.Game()
    p = gm.platforms[3]
    gm.player.rect.midbottom = p.rect.midtop; gm.player.standing_on = p; gm.player.on_ground = True
    offs=set(); xs=[]
    for f in range(400):
        gm.update(K()); offs.add(gm.player.rect.x - p.rect.x); xs.append(p.rect.x)
    print(seed, 'speed', p.speed, 'bounds', p.bounds, 'x range', min(xs), max(xs), 'player offsets', sorted(of
fs)[:5], 'still on', gm.player.standing_on is p)
" 2>&1 | grep -v Hello | grep -v pygame
0 speed -1 bounds (167, 287) x range 167 287 player offsets [41] still on True
1 speed -1 bounds (213, 333) x range 213 333 player offsets [50] still on True
2 speed -1 bounds (117, 237) x range 117 237 player offsets [38] still on True
3 speed -1 bounds (202, 320) x range 202 320 player offsets [62] still on True
4 speed -1 bounds (6, 126) x range 6 126 player offsets [26] still on True
```

The player is carried with a constant offset, so it works. Committing Task 3.

Tool call: Bash Commit Task 3

```
git commit -q -am "Task 3: Implement moving_platform_speed

Every third (non-ground) platform now oscillates horizontally. Speed
ramps from 1 px/frame near the bottom to 3 px/frame at the top, and the
starting direction alternates between consecutive moving platforms.
Speeds are whole numbers because pygame.Rect positions are integers." && git log --oneline | head -1
1dd2cb7 Task 3: Implement moving_platform_speed
```

Task 4: a sparkle burst and a combo counter when a coin is collected.

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
SPARKLE_COUNT, SPARKLE_LIFE, POPUP_LIFE = 14, 30, 45
SPARKLES = []  # live sparkle particles: [x, y, vx, vy, frames_left]
```

```
POPUPS = []  # floating "+50" labels: [x, y, text, frames_left]
COMBO = {"count": 0, "best": 0}  # coins collected in a row without losing a life


def on_coin_collected(coin, score):
    """Called the instant the player collects a coin, after its value has been added to the score. Add a sound
 or sparkle here."""
    COMBO["count"] += 1
    COMBO["best"] = max(COMBO["best"], COMBO["count"])
    for i in range(SPARKLE_COUNT):
        angle = 2 * math.pi * i / SPARKLE_COUNT + random.uniform(-0.2, 0.2)
        speed = random.uniform(1.5, 3.5)
        SPARKLES.append([coin.pos.x, coin.pos.y, math.cos(angle) * speed, math.sin(angle) * speed, SPARKLE_LIF
E])
    label = "+50" if COMBO["count"] < 2 else f"+50  x{COMBO['count']} combo!"
    POPUPS.append([coin.pos.x, coin.pos.y - 12, label, POPUP_LIFE])


def reset_effects():
    SPARKLES.clear()
    POPUPS.clear()
    COMBO["count"] = COMBO["best"] = 0


def update_effects():
    for s in SPARKLES:
        s[0] += s[2]
        s[1] += s[3]
        s[3] += 0.15  # sparkles drift down slightly
        s[4] -= 1
    for p in POPUPS:
        p[1] -= 0.8
        p[3] -= 1
    SPARKLES[:] = [s for s in SPARKLES if s[4] > 0]
    POPUPS[:] = [p for p in POPUPS if p[3] > 0]


def draw_effects(screen, cam_y, font):
    for x, y, _, _, life in SPARKLES:
        fade = life / SPARKLE_LIFE
        color = (255, round(200 + 55 * fade), round(80 + 175 * fade))
        pygame.draw.circle(screen, color, (x, y - cam_y), max(1, round(3 * fade)))
    for x, y, text, life in POPUPS:
        label = font.render(text, True, (255, 230, 120))
        label.set_alpha(round(255 * min(1.0, life / (POPUP_LIFE / 2))))
        screen.blit(label, label.get_rect(center=(x, y - cam_y)))
```

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
import math
import random
import pygame
The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully.
```

The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully.

Now I'll wire the effects into `Game` (reset, update, losing a life, and drawing):

Tool call: Bash Hook effects into Game reset, update, life loss, and draw

```
python3 - <<'EOF'
import re
src = open("game.py").read()
def sub(old, new):
    global src
    assert src.count(old) == 1, old
    src = src.replace(old, new)
sub("""        self.state = "play"
        self.top_y = self.platforms[-1].rect.y
""", """        self.state = "play"
        self.top_y = self.platforms[-1].rect.y
        reset_effects()
""")
sub("""    def update(self, keys):
        if self.state != "play":
            return
""", """    def update(self, keys):
        update_effects()  # let sparkles finish even after the game ends
        if self.state != "play":
            return
""")
sub("""            self.lives -= 1
            if self.lives <= 0:""", """            self.lives -= 1
            COMBO["count"] = 0  # falling breaks the coin combo
            if self.lives <= 0:""")
sub("""        self.player.draw(screen, self.cam_y)

        hud = self.font.render(f"Height: {self.height}m  Coins: {self.coin_score // 50}  Lives: {self.lives}",
 True, (200, 200, 200))
        screen.blit(hud, (10, 10))
```

```
""", """        self.player.draw(screen, self.cam_y)
        draw_effects(screen, self.cam_y, self.font)

        hud = self.font.render(f"Height: {self.height}m  Coins: {self.coin_score // 50}  Lives: {self.lives}",
 True, (200, 200, 200))
        screen.blit(hud, (10, 10))
        combo = self.font.render(f"Combo: x{COMBO['count']}  (best x{COMBO['best']})", True, (255, 210, 90))
        screen.blit(combo, (10, 32))
""")
open("game.py", "w").write(src)
EOF
git diff --stat
game.py | 52 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++-
 1 file changed, 51 insertions(+), 1 deletion(-)
```

Tool call: Bash Test coin effects, combo reset, and lint the code

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && SDL
_VIDEODRIVER=dummy $S/venv/bin/python -W error -c "
import random, pygame, game
pygame.init(); screen = pygame.display.set_mode((game.WIDTH, game.HEIGHT))
class K:
    def __getitem__(s,k): return False
random.seed(1); gm = game.Game()
c = gm.coins[0]
gm.player.rect.center = (int(c.pos.x), int(c.pos.y)); gm.player.vel_y = 0
gm.update(K())
print('after coin: score', gm.coin_score, 'combo', game.COMBO, 'sparkles', len(game.SPARKLES), 'popups', len(g
ame.POPUPS), 'coin removed', c not in gm.coins)
gm.draw(screen)
for _ in range(60): gm.update(K()); gm.draw(screen)
print('after 60 frames sparkles', len(game.SPARKLES), 'popups', len(game.POPUPS))
# fall off bottom
gm.player.rect.y = gm.cam_y + game.HEIGHT + 100
gm.update(K()); print('after fall lives', gm.lives, 'combo', game.COMBO)
gm.reset(); print('after reset combo', game.COMBO, len(game.SPARKLES))
" 2>&1 | grep -v -e Hello -e 'pygame 2'; python3 -m pyflakes game.py 2>/dev/null || $S/venv/bin/pip -q install
 pyflakes && $S/venv/bin/python -m pyflakes game.py && echo pyflakes-clean
after coin: score 50 combo {'count': 1, 'best': 1} sparkles 14 popups 1 coin removed True
after 60 frames sparkles 0 popups 0
after fall lives 2 combo {'count': 0, 'best': 1}
after reset combo {'count': 0, 'best': 0} 0

[notice] A new release of pip is available: 24.2 -> 26.2.1
[notice] To update, run: python3.13 -m pip install --upgrade pip
pyflakes-clean
```

Everything checks out and lint is clean. Recording candidate "after" videos:

Tool call: Bash Record candidate after-change videos across seeds

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; cd $S
 && for s in 1 2 3 4 5 6 7 8 9 10 11 12; do venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4/37_cl
imber/game.py try/a$s.mp4 $s 7 10 2>&1 | grep seed; done
seed=1 peak=96 final=96 lives=3 state=play coins=4 dropped_below_peak=False
seed=2 peak=95 final=95 lives=3 state=play coins=3 dropped_below_peak=False
seed=3 peak=90 final=90 lives=0 state=lose coins=4 dropped_below_peak=False
seed=4 peak=51 final=51 lives=3 state=play coins=2 dropped_below_peak=False
seed=5 peak=98 final=98 lives=3 state=play coins=2 dropped_below_peak=False
seed=6 peak=96 final=96 lives=3 state=play coins=6 dropped_below_peak=False
seed=7 peak=96 final=96 lives=3 state=play coins=3 dropped_below_peak=False
seed=8 peak=103 final=103 lives=3 state=play coins=3 dropped_below_peak=False
seed=9 peak=91 final=91 lives=3 state=play coins=4 dropped_below_peak=False
seed=10 peak=56 final=56 lives=3 state=play coins=0 dropped_below_peak=False
seed=11 peak=106 final=106 lives=3 state=play coins=1 dropped_below_peak=False
seed=12 peak=107 final=107 lives=3 state=play coins=0 dropped_below_peak=False
Shell cwd was reset to /Users/gaurav/Desktop/SE_Labs
```

Seed 6 generates the same level as the "before" video, which makes for a direct comparison. Checking frames:

Tool call: Bash Extract tiled preview frames from after video

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; FF=$(
$S/venv/bin/python -c "import imageio_ffmpeg;print(imageio_ffmpeg.get_ffmpeg_exe())"); for n in 120 160 200 24
0 280 320 360 400 440 480 520 590; do :; done; $FF -loglevel error -y -i $S/try/a6.mp4 -vf "select='eq(n\,150)
+eq(n\,250)+eq(n\,330)+eq(n\,400)+eq(n\,500)+eq(n\,590)',scale=320:-1,tile=3x2" -frames:v 1 $S/try/a6_tile.png
 && echo ok
ok
```

Tool call: Read
/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/try/a6_tile.png

```
[image shown to the assistant]
```

Tool call: Bash Enlarge sparkles and review the Task 4 diff

```
cd /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber && sed -i '' 's/^SPARKLE_COUNT, SPARKLE_LIFE, POPUP_LIFE = 14
, 30, 45$/SPARKLE_COUNT, SPARKLE_LIFE, POPUP_LIFE = 16, 32, 45/; s/max(1, round(3 \* fade))/max(1, round(4 * f
ade))/' game.py && grep -n "SPARKLE_COUNT, SPARKLE_LIFE\|4 \* fade" game.py && git diff | head -120
33:SPARKLE_COUNT, SPARKLE_LIFE, POPUP_LIFE = 16, 32, 45
74:        pygame.draw.circle(screen, color, (x, y - cam_y), max(1, round(4 * fade)))
```

```
diff --git a/game.py b/game.py
index 01f2d8a..d4a5a70 100644
--- a/game.py
+++ b/game.py
@@ -1,3 +1,4 @@
+import math
 import random
 import pygame

@@ -29,9 +30,52 @@ def moving_platform_speed(index, total):
     return speed if (index // 3) % 2 else -speed  # alternate starting direction


+SPARKLE_COUNT, SPARKLE_LIFE, POPUP_LIFE = 16, 32, 45
+SPARKLES = []  # live sparkle particles: [x, y, vx, vy, frames_left]
+POPUPS = []  # floating "+50" labels: [x, y, text, frames_left]
+COMBO = {"count": 0, "best": 0}  # coins collected in a row without losing a life
+
+
 def on_coin_collected(coin, score):
     """Called the instant the player collects a coin, after its value has been added to the score. Add a soun
d or sparkle here."""
-    pass
+    COMBO["count"] += 1
+    COMBO["best"] = max(COMBO["best"], COMBO["count"])
+    for i in range(SPARKLE_COUNT):
+        angle = 2 * math.pi * i / SPARKLE_COUNT + random.uniform(-0.2, 0.2)
+        speed = random.uniform(1.5, 3.5)
+        SPARKLES.append([coin.pos.x, coin.pos.y, math.cos(angle) * speed, math.sin(angle) * speed, SPARKLE_LI
FE])
+    label = "+50" if COMBO["count"] < 2 else f"+50  x{COMBO['count']} combo!"
+    POPUPS.append([coin.pos.x, coin.pos.y - 12, label, POPUP_LIFE])
+
+
+def reset_effects():
+    SPARKLES.clear()
+    POPUPS.clear()
+    COMBO["count"] = COMBO["best"] = 0

... [2270 more characters truncated]
```

I'll keep the combo popups from getting clipped at the screen edges:

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
        rect = label.get_rect(center=(x, y - cam_y))
        rect.x = max(0, min(WIDTH - rect.width, rect.x))  # keep the text on screen near the edges
        screen.blit(label, rect)
```

Tool call: Bash Lint, test, and commit Task 4

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && $S/
venv/bin/python -m pyflakes game.py && (cd $S && venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4/
37_climber/game.py try/t4.mp4 6 7 10 2>&1 | grep seed) && git commit -q -am "Task 4: Implement on_coin_collect
ed with sparkles and a coin combo

Collecting a coin now:
- bursts a ring of sparkle particles from the coin's position that
  fly outward, drift down and fade out
- shows a floating \"+50\" label that rises and fades
- increments a combo counter for coins collected in a row without
  losing a life (shown in the HUD with the best combo of the run)

Losing a life resets the current combo; pressing R clears all effects
and the combo. Effects keep animating after the game ends so the last
burst finishes cleanly." && git log --oneline | head -1
The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully. (note: the file
had been modified on disk since you last read it — the edit applied cleanly, but the file contains other chang
es not in your context. Read it before edits that depend on surrounding content.)
seed=6 peak=96 final=96 lives=3 state=play coins=6 dropped_below_peak=False
554fa21 Task 4: Implement on_coin_collected with sparkles and a coin combo
```

Now the gradient refinement: a multi-stop gradient so the colour change is visible within the first few platforms but still spans the whole climb.

Tool call: Edit /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py

```
    # Fade through green -> teal -> violet -> pink -> gold from the ground to the top of the climb,
    # so the colour shift is already visible a few platforms up.
    stops = [(100, 180, 100), (70, 170, 200), (140, 120, 235), (235, 105, 170), (245, 195, 70)]
    t = max(0.0, min(1.0, index / total)) if total else 0.0
    pos = t * (len(stops) - 1)
    i = min(int(pos), len(stops) - 2)
    frac = pos - i
    return tuple(round(a + (b - a) * frac) for a, b in zip(stops[i], stops[i + 1]))
```

Tool call: Bash Check multi-stop gradient values

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && SDL
_VIDEODRIVER=dummy $S/venv/bin/python -c "
```

```
import game
print([game.platform_color(i,60) for i in (0,5,10,15,30,45,59,60)], game.platform_color(0,0), game.platform_co
lor(70,60))
" | tail -1
The file /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py has been updated successfully.
[(100, 180, 100), (90, 177, 133), (80, 173, 167), (70, 170, 200), (140, 120, 235), (235, 105, 170), (244, 189,
 77), (245, 195, 70)] (100, 180, 100) (245, 195, 70)
```

Now I'll fix the recorder so the climber reliably walks off its platform at the end, then record the final "after" clip:

Tool call: Edit
/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/record.py

```
    if frame >= g.FPS * climb_s:  # walk off the edge to fall back down
        if autopilot.drop_key is None:
            plat = p.standing_on or autopilot.last
            autopilot.drop_key = pygame.K_LEFT if plat.rect.left > g.PLAYER_W + 4 else pygame.K_RIGHT
        pressed.add(autopilot.drop_key)
        return Keys(pressed)
The file /private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad
/record.py has been updated successfully. (file state is current in your context — no need to Read it back)
```

Tool call: Bash Record after videos with the fixed drop autopilot

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; sed -
i '' 's/^autopilot.last = None$/autopilot.last = None\nautopilot.drop_key = None/' $S/record.py && cd $S && fo
r s in 6 1 2 9; do venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py try/f$s.mp4
 $s 7.5 10 2>&1 | grep seed; done; echo ---; venv/bin/python record.py try_orig.py x.mp4 6 2>/dev/null | true
seed=6 peak=108 final=108 lives=0 state=lose coins=5 dropped_below_peak=False
seed=1 peak=110 final=110 lives=0 state=lose coins=4 dropped_below_peak=False
seed=2 peak=102 final=102 lives=0 state=lose coins=3 dropped_below_peak=False
seed=9 peak=91 final=91 lives=0 state=lose coins=4 dropped_below_peak=False
---
Shell cwd was reset to /Users/gaurav/Desktop/SE_Labs
```

The autopilot now walks off every time it respawns and loses all three lives. I'll make it stop after one fall:

Tool call: Bash Make autopilot stop after one fall and re-record

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; cd $S
 && python3 - <<'EOF'
src = open("record.py").read()
old = """        pressed.add(autopilot.drop_key)
        return Keys(pressed)"""
new = """        if not p.on_ground:
            autopilot.airborne = True
        elif autopilot.airborne:  # landed again (lower platform or respawn): stop
            autopilot.drop_key = 0
        if autopilot.drop_key:
            pressed.add(autopilot.drop_key)
        return Keys(pressed)"""
assert src.count(old) == 1
src = src.replace(old, new).replace("autopilot.drop_key = None\n", "autopilot.drop_key = None\nautopilot.airbo
rne = False\n")
open("record.py", "w").write(src)
EOF
for s in 6 1 2 9 3 5; do venv/bin/python record.py /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber/game.py try/f
$s.mp4 $s 7.5 10 2>&1 | grep seed; done
seed=6 peak=108 final=108 lives=2 state=play coins=6 dropped_below_peak=False
seed=1 peak=110 final=110 lives=3 state=play coins=4 dropped_below_peak=False
seed=2 peak=102 final=102 lives=2 state=play coins=3 dropped_below_peak=False
seed=9 peak=91 final=91 lives=3 state=play coins=4 dropped_below_peak=False
seed=3 peak=90 final=90 lives=2 state=play coins=4 dropped_below_peak=False
seed=5 peak=98 final=98 lives=3 state=play coins=2 dropped_below_peak=False
Shell cwd was reset to /Users/gaurav/Desktop/SE_Labs
```

Tool call: Bash Extract preview frames from final after candidates

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad; FF=$(
$S/venv/bin/python -c "import imageio_ffmpeg;print(imageio_ffmpeg.get_ffmpeg_exe())"); for s in 1 6; do $FF -l
oglevel error -y -i $S/try/f$s.mp4 -vf "select='eq(n\,200)+eq(n\,380)+eq(n\,449)+eq(n\,500)+eq(n\,540)+eq(n\,5
99)',scale=320:-1,tile=6x1" -frames:v 1 $S/try/f${s}_tile.png; done; echo ok
ok
```

Tool call: Read
/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/try/f6_tile.png

```
[image shown to the assistant]
```

Tool call: Read
/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad/try/f1_tile.png

```
[image shown to the assistant]
```

Tool call: Bash Commit gradient refinement and save after video

```
cd /Users/gaurav/Desktop/SE_Labs/Lab4/37_climber && S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/81
76d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && $S/venv/bin/python -m pyflakes game.py && git commit -q -am "
Task 2: Use a multi-stop colour gradient for platforms

The two-colour fade was hard to notice over the first dozen platforms.
Platforms now pass through green -> teal -> violet -> pink -> gold from
```

```
the ground to the top, so the change is visible early in the climb
while still shifting gradually as index approaches total." && git log --oneline && cp $S/try/f6.mp4 ../Gaurav_
Girish_PES1UG24AM103/after_changes.mp4
fd08095 Task 2: Use a multi-stop colour gradient for platforms
554fa21 Task 4: Implement on_coin_collected with sparkles and a coin combo
1dd2cb7 Task 3: Implement moving_platform_speed
156ccb1 Task 2: Implement platform_color as a height-based gradient
6313e9f Task 1: Fix height score so it tracks the best height reached
63be6df edit readme for submission reqs
ad335a8 push climber
```

Now a stress test: random inputs across many seeds, plus forcing a win, a loss and resets, to catch any crashes:

Tool call: Bash Fuzz-test the game with random inputs across seeds

```
S=/private/tmp/claude-501/-Users-gaurav-Desktop-SE-Labs/8176d5b2-2142-4eec-aef2-1a46ce2f11b9/scratchpad && SDL
_VIDEODRIVER=dummy $S/venv/bin/python -W error -c "
import random, pygame, game
pygame.init(); screen = pygame.display.set_mode((game.WIDTH, game.HEIGHT))
keys=[pygame.K_LEFT,pygame.K_RIGHT,pygame.K_SPACE,pygame.K_a,pygame.K_d,pygame.K_w,pygame.K_UP]
class K:
    def __init__(s,p): s.p=p
    def __getitem__(s,k): return k in s.p
states=set()
for seed in range(30):
    random.seed(seed); gm=game.Game(); prev=0
    for f in range(3000):
        gm.update(K({k for k in keys if random.random()<0.3})); gm.draw(screen)
        assert gm.height >= prev; prev = gm.height
        if f==1500 and seed%3==0:  # force a win
            gm.player.rect.y = gm.top_y - 5
    states.add(gm.state); gm.reset(); assert gm.height==0 and game.COMBO=={'count':0,'best':0}
print('ok, end states seen:', states)
" 2>&1 | grep -v -e Hello -e 'pygame 2'; git status --short
ok, end states seen: {'win', 'play', 'lose'}
```

No crashes across 30 seeds and 90,000 frames, and the height never decreased. I'll add a short "Changes Made" section to the README as its own commit:

Tool call: Bash Add Changes Made section to README and commit

```
python3 - <<'EOF'
src = open("README.md").read()
old = "---\n\n## Folder Structure"
assert src.count(old) == 1
new = """---

## Changes Made

| Task | Change |
|------|--------|
| 1. Height-score bug | `Game.update` now keeps `self.height = max(self.height, current_height)`, so the HUD s
hows the best height of the run and never drops when the player falls or rides back down. `reset()` still sets
 it to 0. |
| 2. `platform_color` | Platforms fade through green → teal → violet → pink → gold as `index` approaches `tota
l`; the ground keeps the default green. |
| 3. `moving_platform_speed` | Every third platform moves. Speed ramps from 1 to 3 px/frame with height, and t
he starting direction alternates. |
| 4. `on_coin_collected` | Collecting a coin bursts sparkle particles, shows a floating "+50" label, and incre
ments a combo counter (coins in a row without losing a life). The HUD shows the current and best combo. Losing
 a life resets the combo, and `R` clears everything. |

---

## Folder Structure"""
open("README.md", "w").write(src.replace(old, new))
EOF
git commit -q -am "Document the bug fix and implemented features in README" && git log --oneline | head -1
1b202d1 Document the bug fix and implemented features in README
```

Pushing to a new public repo on your personal GitHub account (assignment step 9; no PR to SETAPESU26):

Tool call: Bash Create personal GitHub repo and push commits

```
gh repo view gaurav-2030/37_climber >/dev/null 2>&1 && echo EXISTS || (gh repo create gaurav-2030/37_climber -
-public --description "Climber Repair Lab (SE Lab 4 - VibeCoding): height-score fix, platform colours, moving
platforms, coin sparkles & combo" --source . --remote origin --push 2>&1 | tail -3); git remote -v; git log --
format='%h %an <%ae> %s' origin/main 2>/dev/null | head -8
To https://github.com/gaurav-2030/37_climber.git
 * [new branch]      HEAD -> main
branch 'main' set up to track 'origin/main'.
origin■https://github.com/gaurav-2030/37_climber.git (fetch)
origin■https://github.com/gaurav-2030/37_climber.git (push)
upstream■https://github.com/SETAPESU26/37_climber (fetch)
upstream■https://github.com/SETAPESU26/37_climber (push)
1b202d1 Gaurav Girish <gaurav.girish2006@gmail.com> Document the bug fix and implemented features in README
fd08095 Gaurav Girish <gaurav.girish2006@gmail.com> Task 2: Use a multi-stop colour gradient for platforms
554fa21 Gaurav Girish <gaurav.girish2006@gmail.com> Task 4: Implement on_coin_collected with sparkles and a co
in combo
```

```
1dd2cb7 Gaurav Girish <gaurav.girish2006@gmail.com> Task 3: Implement moving_platform_speed
156ccb1 Gaurav Girish <gaurav.girish2006@gmail.com> Task 2: Implement platform_color as a height-based gradien
t
6313e9f Gaurav Girish <gaurav.girish2006@gmail.com> Task 1: Fix height score so it tracks the best height reac
hed
63be6df Vishwambhara R Hebbalalu <vishwambharah@gmail.com> edit readme for submission reqs
ad335a8 Vishwambhara R Hebbalalu <vishwambharah@gmail.com> push climber
```

The code is pushed. Next, exporting this chat as the chat-history PDF:

Tool call: ToolSearch